
Calculators and communication devices are not allowed.

Answer the following questions.

1. (a) (10 pts.) Prove that a matrix A is invertible if and only if, $\det(A) \neq 0$.
(b) (10 pts.) Let A and B be 3×3 matrices, where $\det(A) = -1$ and $\det(B) = 2$. Find $\det(-2A^T B^{-2} \text{adj}(B))$.
(c) (10 pts.) Determine whether the vectors $X_1 = (3, 0, 0, 0)$, $X_2 = (0, -1, 2, 1)$, $X_3 = (6, 2, -6, 0)$, and $X_4 = (3, -2, 3, 3)$ span the vector space $\mathbb{R}^4$.
2. (a) (10 pts.) Let U and V be unit vectors in $\mathbb{R}^n$. Prove that $\|U + V\|^2 - \|U - V\|^2 \leq 4$.
(b) Let $X = (2, 0, 6, 0, -3)$ and $Y = (2, 1, 3, 1, 1)$ be vectors in $\mathbb{R}^5$.
(i) (10 pts.) Find a unit vector in the direction of Y .
(ii) (10 pts.) Evaluate $\|-2\|X\|Y\|$.
3. (a) (10 pts.) Show that the solution set of a homogeneous linear system $AX = O$ in n unknowns is a subspace of $\mathbb{R}^n$.
(b) Let V be a vector space with a zero vector O . Show that:
(i) (10 pts.) the zero vector O of V is unique.
(ii) (10 pts.) $0X = O$ for any vector X in V .
4. In each of the following questions, select the correct answer. (2 pts. each)
(a) Let U and V be vectors in $\mathbb{R}^5$, such that $\|U - V\| = 3$, then
(A) $\|U - V\| = -3$
(B) $\|U\| = 3 - \|V\|$
(C) $\|U\| = 3 + \|V\|$
(D) None of the above
(b) Let A be a 3×3 matrix. If $A^2 \text{adj}(A) = O$, then
(A) $A = O$
(B) $\text{adj}(A) = O$
(C) $A = O$ or $\det(A) = 0$
(D) All of the above
(c) Let X and Y be vectors in $\mathbb{R}^n$, then $\|X + Y\|^2 =$
(A) $X^2 + 2XY + Y^2$
(B) $\|X\|^2 + 2\|X\|\|Y\| + \|Y\|^2$
(C) $\|X\|^2 + 2X \cdot Y + \|Y\|^2$
(D) All of the above
(d) Let A and B be 4×4 matrices, then
(A) $\det(A + AB) = \det(A + BA)$
(B) $\det(AB + B) = \det(A + BA)$
(C) $\det(A + B) = \det(A) + \det(B)$
(D) None of the above
(e) Let X and Y be linearly independent vectors in a vector space V . Then
(A) $X = cY$ for a nonzero scalar c .
(B) $\{O, X, Y\}$ is linearly independent
(C) kX and kY are linearly independent for any nonzero scalar k
(D) None of the above

1. (a) (10 pts.) (I) A is invertible $\implies A^{-1}$ exists, such that $AA^{-1} = I_n \implies |AA^{-1}| = |I_n| \implies |A| |A^{-1}| = |I_n| = 1 \implies |A| \neq 0$.
 (II) $\because \text{adj}(A)A = |A|I_n$. If $|A| \neq 0$, then $\frac{1}{|A|}\text{adj}(A)A = I_n \implies \frac{1}{|A|}\text{adj}(A) = A^{-1}$ (inverse of A is unique) $\implies A$ is invertible.
- (b) (10 pts.) $\det(-2A^T B^{-2} \text{adj}(B)) = (-2)^3 |A^T| |B^{-2}| |\text{adj}(B)|$
 $= (-2)^3 |A| (|B^{-1}|)^2 |\text{adj}(B)| = (-2)^3 (-1) \left(\frac{1}{2}\right)^2 |B|^{3-1} = 2(2)^2 = 8$.
- (c) (10 pts.) Since $\begin{vmatrix} 3 & 0 & 6 & 3 \\ 0 & -1 & 2 & -2 \\ 0 & 2 & -6 & 3 \\ 0 & 1 & 0 & 3 \end{vmatrix} = -3 \begin{vmatrix} 1 & -2 & 2 \\ 2 & -6 & 3 \\ 1 & 0 & 3 \end{vmatrix} = -3 \begin{vmatrix} 1 & -2 & 2 \\ 0 & -2 & -1 \\ 0 & 2 & 1 \end{vmatrix} =$
 $0 \implies S = \{X_1, X_2, X_3, X_4\}$ does not span $\mathbb{R}^4$.
2. (a) (10 pts.) $\|U + V\|^2 - \|U - V\|^2 = (U + V) \cdot (U + V) - (U - V) \cdot (U - V)$
 $= 2U \cdot V + 2U \cdot V = 4U \cdot V \leq 4\|U\| \|V\| = 4$.
- (b) (i) (10 pts.) $\|Y\| = \sqrt{2^2 + 1^2 + 3^2 + 1^2 + 1^2} = 4 \implies$ Unit vector in the direction of Y , is $U = \frac{1}{\|Y\|}Y = \frac{1}{4}(2, 1, 3, 1, 1) = \left(\frac{1}{2}, \frac{1}{4}, \frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right)$.
 (ii) (10 pts.) $\|X\| = \sqrt{2^2 + 0^2 + 6^2 + 0^2 + (-3)^2} = 7 \implies \|-2\|X\|Y\| = \|-2\|X\| \|Y\| = (2)(7)(4) = 56$
3. (a) Let $W = \{X \in \mathbb{R}^n : AX = O_{m \times 1}\}$. Since $A_{m \times n} O_{n \times 1} = O_{m \times 1}$
 $\implies O_{n \times 1} \in W \implies W \neq \emptyset$.
 (α) Let $X_1, X_2 \in W \implies AX_1 = O_{m \times 1}$ and $AX_2 = O_{m \times 1}$.
 Since $A(X_1 + X_2) = AX_1 + AX_2 = O_{m \times 1} + O_{m \times 1} = O_{m \times 1} \implies X_1 + X_2 \in W$
 (β) For any $k \in \mathbb{R}$, and any $X \in W$ (then $AX = O_{m \times 1}$), we have $A(kX) = k(AX) = kO_{m \times 1} = O_{m \times 1} \implies kX \in W$.
 Then W is a subspace of $\mathbb{R}^n$.
- (b) (i) (10 pts.) Assume O_1 and O_2 are zero vectors in V . $\implies O_1 = O_1 + O_2$ (O_2 is a zero vector) $= O_2$ (O_1 is a zero vector). $\implies O_1 = O_2$
 (ii) (10 pts.) $0X = (0 + 0)X = (0X + 0X) \implies \underline{0} = 0X + (-0X)$
 $= (0X + 0X) + (-0X) = (0X + (-0X)) + 0X = O + 0X = \underline{0X} \implies 0X = O$.
4. (a) Let U and V be vectors in $\mathbb{R}^5$, such that $\|U - V\| = 3$, then
 (D) None of the above
- (b) Let A be a 3×3 matrix. If $A^2 \text{adj}(A) = O$, then
 (C) $A = O$ or $\det(A) = 0$
- (c) Let X and Y be vectors in $\mathbb{R}^n$, then $\|X + Y\|^2 =$
 (C) $\|X\|^2 + 2X \cdot Y + \|Y\|^2$
- (d) Let A and B be 4×4 matrices, then
 (A) $\det(A + AB) = \det(A + BA)$
- (e) Let X and Y be linearly independent vectors in a vector space V . Then
 (C) kX and kY are linearly independent for any nonzero scalar k